C9 Group 2, the Alkaline Earth Metals – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Trends **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Trend in atomic radius down Group 2				Increases
2	Trend in first IE down Group 2				Decreases
3	Trend in melting point down Group 2				Decreases (generally)
4	Group 2 element with anomalously low melting point				Mg
5	Trend in reactivity with water down Group 2				Increases
6	Trend in solubility of hydroxides down Group 2				Increases
7	Trend in solubility of sulfates down Group 2				Decreases
8	Least soluble Group 2 hydroxide				$Mg(OH)_2$
9	Least soluble Group 2 sulfate				$BaSO_4$
10	Ion formed by all Group 2 metals				M^{2+}

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Reactions **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Products of Mg with steam				$MgO + H_2$
2	Equation for Mg with steam				$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$
3	Observation when Mg burns in steam				Bright white light, white solid
4	Products of Mg with cold water				$Mg(OH)_2 + H_2$ (very slow)
5	Products of Ca with water				$Ca(OH)_2 + H_2$
6	Equation for Ca with water				$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$
7	pH of solution formed by Ca with water				Alkaline (about 12)
8	Change in oxidation state of Mg when it reacts				$0 \rightarrow +2$
9	Metal used to extract titanium				Mg
10	Equation for titanium extraction				$TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$

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AQA A-level Chemistry • Set 3: Uses and the Sulfate Test **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Use of $Mg(OH)_2$				Antacid
2	Use of $Ca(OH)_2$				Neutralising acidic soil
3	Use of CaO or $CaCO_3$				Removing SO_2 from flue gases
4	Equation for CaO removing SO_2				$CaO + SO_2 \rightarrow CaSO_3$
5	Use of $BaSO_4$				Barium meal for X-rays
6	Why $BaSO_4$ is safe despite barium being toxic				Insoluble
7	Reagent to test for sulfate ions				Acidified $BaCl_2$
8	Positive result for sulfate ions				White precipitate
9	Why the $BaCl_2$ is acidified				Removes CO_3^{2-} so white $BaCO_3$ cannot form
10	Ionic equation for the sulfate test				$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

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AQA A-level Chemistry • Set 4: Equations **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Equation for Mg with cold water				$Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$
2	Equation for Sr with water				$Sr + 2H_2O \rightarrow Sr(OH)_2 + H_2$
3	Equation for Ba with water				$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$
4	General equation for a Group 2 metal M with water				$M + 2H_2O \rightarrow M(OH)_2 + H_2$
5	Equation for Mg burning in oxygen				$2Mg + O_2 \rightarrow 2MgO$
6	Equation for $Mg(OH)_2$ as an antacid				$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$
7	Ionic equation for $Ca(OH)_2$ neutralising acidic soil				$Ca(OH)_2 + 2H^+ \rightarrow Ca^{2+} + 2H_2O$
8	Equation for $CaCO_3$ removing SO_2				$CaCO_3 + SO_2 \rightarrow CaSO_3 + CO_2$
9	Ionic equation for Mg^{2+} with OH^-				$Mg^{2+} + 2OH^- \rightarrow Mg(OH)_2$
10	Ionic equation for the false positive in the sulfate test				$Ba^{2+} + CO_3^{2-} \rightarrow BaCO_3$

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